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# Computational and experimental exploration of a morphable blade concept for wind turbines

Stephane Redonnet *, Antoine Bach, Sahil Kunder, Thomas Schmidt, Nazmus Rahman

*The Hong Kong University of Science & Technology, Department of Mechanical & Aerospace Engineering, Hong Kong Special Administrative Region of China*

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ABSTRACT

Regarding the further development of wind energy solutions, this study explores the idea of morphing wind turbine blades as a means to improve their aero-structural merits. To this end, the aerodynamics of a representative, morphable wind turbine blade section is assessed using numerical simulation, this being done for a significant range of flow conditions (speed, incidence), with various levels of morphing applied. Upon this, diverse morphing scenarios are inferred, either to alleviate the aero-structural loads exerted on the blade, or to rather maximize its generated aerodynamic power. It is shown that these morphing strategies translate into significant benefits, among which substantial gains of power delivered across all flow regimes. The phenomenology behind the benefits brought by the morphing is then explored, which confirms that morphing a blade is superior to simply pitching or twisting it – as classically done or envisioned. Finally, the outcomes obtained from the computational investigation are replicated through a dedicated, small-scale experiment, thereby further confirming the aerodynamic merits brought by the morphing. All this advocates for a further exploration of morphable wind turbine blades.

## 1. Introduction

Following a century of intensive exploitation of fossil fuels, massive efforts are now channeled into developing technologies that rely on non-fossil forms of energy supply, whether alternative (e.g., natural gas and fuel cells) or renewable (Renewables 2018; Global Status, 2018) (e.g., solar, wind and tides). One of the most promising means is the use of wind-generated energy, which requires the further development and massive deployment of on- and/or offshore wind farms. However, the efforts for boosting capacities in wind energy should not only be massive, but smart as well: Beyond just relying on the further deployment of larger wind farms, one should also seek to increase the power generation at the level of wind turbines, by making them more efficient. Notably, the efficiency of a wind turbine is essentially driven by the aero-structural characteristics of its blades. While the power produced is a function of the aerodynamic performances by the blades (e.g., maximum lift and/or lift-to-drag ratio), it is also bounded by its structural limits (e.g., critical loads entailed by excessive lift). The point is that wind turbines usually undergo a wide range of operating conditions, depending on the wind force and the subsequent rotational speed of their rotor. Consequently, the blades may experience very diverse airflow conditions (e.g., flow speed and incidence) which may question their aerodynamic efficiency and – sometimes – jeopardize their structural integrity. In other words, the performances by wind turbine blades can be poor when the operating conditions are not optimal, either due to losses in their aerodynamic efficiency (e.g., stall phenomena) or because of their structural limitations (e.g., excessive aerodynamic loads).

To tentatively overcome these issues, most modern wind turbines are equipped with specific devices, which aim at tailoring the blades' aerodynamic efficiency to different wind conditions (Muljadi and Butterfield, 1999). For instance, pitch adjustment systems allow modifying the blades' angle of attack, so as to increase their lift-to-drag ratio when the power is too small, or – oppositely – to decrease their lift when the structural loads may be too high. These mechanisms, however, are of limited efficiency, whether because they are not versatile enough for optimizing the aerodynamic characteristics at each blade section (global pitch, with only a few presets available), or because they may not be sufficient to limit the structural loads on some parts of the blade – which may impose to temporarily stop the wind turbine. Other mitigation systems exist, such as retractable appendices which are implemented on the blades themselves, like flaps (Xu et al., 2018; Oltmann et al., 2017; Kentfield, 1994; Bianchini et al., 2019; Nikoueeyan et al., 2015). These devices, however, offer mixed benefits (Xu et al., 2018; Oltmann et al., 2017; Kentfield, 1994). For instance, traditional flaps proved to reduce

* Corresponding author.
*E-mail address:* redonnet@ust.hk (S. Redonnet).

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the bending moment at the blades' root if used instead of a pitch adjustment mechanism (Oltmann et al., 2017), but they incur extra losses in aerodynamic efficiency (Xu et al., 2018). Comparatively, Gurney flaps were shown to degrade less the blades' aerodynamic performances (Xu et al., 2018), and – sometimes – to even improve them (Kentfield, 1994). However, these flap-based solutions must often be used in conjunction with pitch adjustment mechanisms (Lackner and van Kuik, 2010; Unguran and Kühn, 2016), which may overcomplicate the design, the production (manufacturing) and the operation (control) of wind turbine blades. Other technological solutions were – and are still – explored (e.g., vortex-generators, micro-tabs, blowing/suction devices, plasma actuators), which all come with their own advantages and drawbacks. As a result, a universal, consensual means for enhancing the wind turbines efficiency via a perfectly tailored control of their blade aerodynamics is yet to be developed.

Over the past two decades, various studies investigated how morphing mechanisms could offer deforming on demand the trailing edge section of aircraft wings and, more recently, wind turbine blades. Two main categories of designs have emerged, namely compliant (Dayyani et al., 2014; Lachenal et al., 2013) and kinematic (Monner et al., 2000; Arena et al., 2019; Karakalas et al., 2016) mechanisms. Whereas the former consist of a fixed internal structure that can deform to deflect the trailing edge section, the latter rather rely on a mechanical assembly of actuated parts that are moved (sometimes independently from each other) so as to enforce a similar deflection of the trailing edge. Both types of design share the same requirement of a flexible outer skin, which must ensure that the morphable wing (or blade) exhibits a smooth profile. On that stage, diverse solutions were explored, such as elastomeric materials (Arena et al., 2019) as well as metallic compliant structures (You et al., 2019; Molinari et al., 2016). In recent years, various research efforts explored how morphing concepts could be applied to wind turbine blades (Dayyani et al., 2014; Lachenal et al., 2013; Karakalas et al., 2016; Aia et al., 2019; Shroff et al., 2016; Boorsma and Schepers, 2015; Frederik et al., 2017). In particular, one can cite the European Project INNWIND.EU, whose objective was to design novel wind turbines of improved efficiency (Karakalas et al., 2016; Shroff et al., 2016; Boorsma and Schepers, 2015; Frederik et al., 2017). Within this framework, preliminary studies were conducted, which aimed at exploring a morphable blade concept (Karakalas et al., 2016; Frederik et al., 2017). These pioneering works primarily focused on the structural aspects of the design, which were assessed through dedicated loads tests (Frederik et al., 2017) with only basic information on the aerodynamics obtained experimentally.

The present study aims at building upon that pioneering exploration while going a step further, by assessing the proposed morphable blade concept from an aerodynamic efficiency perspective. To this end, the generic section of a representative wind turbine blade is morphed, its aerodynamic performances being characterized using both computational and experimental means. A sufficient number of flow conditions are explored to make the observations representative enough of the typical operational conditions one may expect for classical wind turbines. Besides, a representative number of morphed configurations are tackled, to derive an optimal morphing strategy tailorable to any section along the blade span, depending on the local characteristics (geometry, flow conditions). All in all, more than 160 simulations are performed using Computational Fluid Dynamics, a fraction of them being then validated using small scale experiments.

The paper is organized as follows; Section 1 introduces the rationale and strategy underlying such an exploration of morphable wind turbine blades. Section 2 summarizes the computational approach and set-up employed to do so, whereas the numerical results are discussed in Section 3. The latter are validated using a small-scale experiment, whose apparatus and outcomes are described in Section 4. Finally, some conclusions and perspectives are drawn.

## 2. Rationale and strategy for morphable wind turbine blades

The present section first recalls some background knowledge about the aerodynamics of wind turbine blades. It then introduces the morphing strategy that is here explored for enhancing their aerodynamic capacity and power efficiency.

### *2.1. Performances of wind turbines and aerodynamics of their blades*

Wind turbines extract power from the wind through the rotation of their blades, whose motion results from the aerodynamic forces (lift and drag) they are exerted by the airflow (see Fig. 1).

Let us consider a wind turbine blade of radius $R$ that rotates at an angular speed ($\omega$) due to an incoming flow of freestream velocity ($\boldsymbol{v}_\infty$). The incoming flow speed (resp. blade's rotational speed) can be more universally characterized through the so-called Reynolds number, $Re_\infty$ (resp. Tip-Speed-Ratio, $TSR$). Taking the blade radius ($R$) and wind speed ($v_\infty$) as characteristic quantities, these dimensionless parameters are respectively given by

$$Re_\infty \stackrel{\text{def}}{=} \frac{\rho v_\infty R}{\mu} \tag{1}$$

and

$$TSR \stackrel{\text{def}}{=} \frac{R\omega}{v_\infty} \tag{2}$$

with $\rho$ and $\mu$ standing for the fluid density and viscosity, respectively. As detailed in Appendix A, under these conditions, any given section located along the span is to extract a specific aerodynamic power, whose dimensionless coefficient is defined by

$$C_P(\alpha,\lambda) \; = \; C_L(\alpha)\left\{\lambda\sqrt{1+\lambda^2}\left[1-\frac{C_D(\alpha)}{C_L(\alpha)}\lambda\right]\right\} \tag{3}$$

In the above equation, $\lambda$ stands for the advancement-to-wind speeds ratio, which describes how fast the blade section travels with respect to the incoming flow velocity ($v_\infty$). By introducing the section's radius ($r$), this ratio can be re-expressed as (cf. Appendix A)

$$\lambda = \frac{r}{R}TSR \tag{4}$$

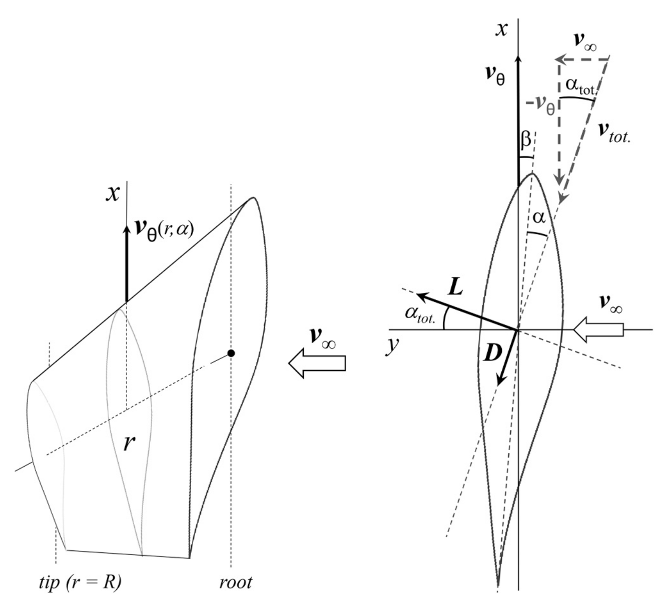


**Fig. 1.** Wind turbine blade aerodynamics. Sketch of a blade (left) and its section (right), with the aerodynamic forces ($\boldsymbol{L}$: lift, $\boldsymbol{D}$: drag) induced by the airflow ($\boldsymbol{v}_{tot.}$) resulting from the wind ($\boldsymbol{v}_\infty$) and rotational ($\boldsymbol{v}_\theta$) velocities.

Besides, in Eq. (3), $C_L$ and $C_D$ respectively stand for the lift and drag coefficients exhibited by the blade section. These coefficients are driven by the section's angle of attack ($\alpha$), which results from how the incoming airflow orientation ($\alpha_{tot.}$) combines with the blade's overall pitch ($\beta_b$) and local twist ($\beta_t$) angles (cf. Appendix A). Aside the angle of attack ($\alpha$), the aerodynamic coefficients ($C_L, C_D$) also depend on the section's *local* Reynolds number ($Re$), which is driven by the local airflow speed ($v_{tot.}$) and airfoil chord length ($c$). Of note, both the airflow speed ($v_{tot.}$) and incidence ($\alpha_{tot.}$) can be expressed in terms of the advancement-to-wind speeds ratio ($\lambda$), namely (cf. Appendix A)

$$v_{tot.} = v_\infty \sqrt{1+\lambda^2} \tag{5}$$

and

$$\alpha_{tot.} = \mathrm{Atan}\left(\frac{1}{\lambda}\right) \tag{6}$$

Using this, one can express the blade section's local Reynolds number as

$$Re = \sqrt{1+\lambda^2}\left(\frac{c}{R}\right)Re_\infty \tag{7}$$

In summary, the blade section exhibits a power coefficient ($C_P$) that is proportional to the lift coefficient ($C_L$), with a proportionality factor which depends on both the advancement-to-wind speeds ratio ($\lambda$) and the lift-to-drag ratio ($C_L/C_D$). The latter quantities are respectively driven by the ratio $r/R$ (through the $\lambda \longleftrightarrow TSR$ relationship, cf. Eq. (4)) and $c/R$ (through the $Re \longleftrightarrow Re_\infty$ relationship, see Eq. (7)).

Table 1 hereafter exemplifies what would be some representative values of these parameters for various sections of a typical NREL Phase VI wind turbine blade ($R \sim 5.5$ m, hub included), the latter being considered under two specific conditions of incoming flow velocity (i.e., $Re_\infty$) and subsequent rotational speed ($TSR$), with specific pitch angles applied each time. These examples illustrate well how each blade section may see its aerodynamic characteristics varying importantly, depending on the wind force and subsequent rotational speed – even though the blade is applied some pitch. For instance, here, the local Reynolds numbers range from as low as 260,000 to almost 2 million, whereas the local incidences ($\alpha$) vary from about 1° to almost 20°. Notably, these significant variations do not only occur across the blade span but may also happen at the level of each section. For instance, between these two operational conditions, $S_4$ section sees its Reynolds number varying from as low as $0.3\times10^6$ to nearly $2\times10^6$, with an incidence ranging from 6° to about 11°. These numbers are indicative only, since other wind forces would translate into different values. They however illustrate well the large variability that a blade (and each one of its sections) may experience, in terms of aerodynamic characteristics – and, thus, expected power delivery.

Therefore, it is deemed impossible to optimize the design of wind turbine blades once for all, i.e., a given design cannot prove optimal across the entire range of operating conditions. As a corollary, this advocates for more versatile means to adjust the blades' aerodynamics, especially if this can be done *locally* (at the level of each section) rather than globally (e.g., through some pitch). This is what motivated the present study, which explores the aerodynamic gain to be possibly offered by a proper morphing of wind turbine blades. To do so, a modified (morphable) version of a representative NREL Phase VI wind turbine blade section is explored computationally, this being done for a representative set of flow velocities ($Re = 0.25\times10^6$ to $2\times10^6$) and incidences ($\alpha = -2.5°$ to 15°). As said, the present morphing strategy relies on the pre-existing solution proposed in INNWIND.EU, namely a morphable trailing edge (Frederik et al., 2017; Karakalas et al., 2016).

### 2.2. Adaptive flow control via trailing edge morphing

The morphing of wing or blade sections has been investigated for two decades, firstly in the aerospace domain, e.g., the Adaptive Compliant Trailing Edge technology developed by Flexsys and explored by NASA. Over the years, various morphing concepts were proposed, each one coming with its own advantages and drawbacks. The most popular solution is to incorporate to the wing/blade section a trailing edge that can smoothly deflect towards the pressure or suction side (Monner et al., 2000), thereby modifying its camber on demand. For instance, Arena *et al.* designed a morphable trailing edge relying on finger-like ribs, each rib being composed of several elements that are articulated through hinges (Arena et al., 2019). Dayyani *et al.* proposed the so-called Fish-BAC design (Dayyani et al., 2014), which resembles the skeleton of a fish, with a compliant strip running along the airfoil camber line and skeleton-like stringers connecting it to an elastomeric skin. This concept allows morphing the downstream section of the airfoil (i.e., from its half-chord to its trailing edge) by up to ±10°. Similarly, Lachenal *et al.* investigated a design built around a central laminate with a flexible honeycomb core, connecting it to a silicone skin (Lachenal et al., 2013). Alternative concepts were proposed, which rely on integrating piezo-electric actuators within a compliant web so as to deform the entire airfoil instead of just the trailing edge (Fusi et al., 2018). For instance, Lachenal *et al.* explored a design relying on air pockets that are located on the airfoil extrados and can be inflated / deflated so as to adjust the camber (Lachenal et al., 2013).

The present study is based on the so-called finger morphing concept, which relies on a trailing edge section made of several movable elements that are connected via hinges and can be actuated independently one from the others (Monner et al., 2000; Karakalas et al., 2016). Despite its relative simplicity, this design allows morphing the downstream half of the airfoil into a wide range of combinations. Such versatility offers an indubitable advantage for an application to wind turbine blades, whose trailing edge can thus be *variably* morphed along the span in order to adapt to the *local* flow characteristics of each section. To assess this, the concept is here employed to design a morphable version of a representative wind turbine blade profile, namely the S809 airfoil. Indeed, since it exhibits superior aerodynamic performances compared to more classical foils (e.g., NACA), this particular profile was widely adopted and extensively documented within the wind turbine research community (Hand et al., 2001; Somers, 1997; Wolfe and Ochs, 1997). For instance, its root part excepted, the NREL Phase VI blade is exclusively based upon this S809 profile, whose generic character makes it particularly interesting, here; The observations made for a given 2D section can be readily transposed to any other section along the span, thereby allowing an easy extrapolation to a full 3D blade.

Here, the S809 airfoil's morphable version incorporates a finger-type trailing edge, which is made of three movable elements capable of

**Table 1**
Main characteristics of five sections ($S_1$ to $S_5$) distributed along the span of a NREL Phase VI wind turbine blade (Hand et al., 2001; Somers, 1997), when considered under a couple of wind speeds ($v_\infty$) and subsequent angular frequencies ($\omega$).

| | | | | $Re_\infty = 1.89\times10^6$ ($v_\infty = 5$ m/s), $TSR = 2.31$ (20 RPM), $\beta_b = 25°$ | | | | | $Re_\infty = 7.57\times10^6$ ($v_\infty = 20$ m/s), $TSR = 4.14$ (143.2 RPM), $\beta_b = 15°$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $r$ (m) | $c$ (m) | $\beta_t$ (°) | $\lambda$ | $v_{tot}$(m/s) | $Re$ | $\alpha_{tot.}$ (°) | $\alpha$(°) | $\lambda$ | $v_{tot}$(m/s) | $Re$ | $\alpha_{tot.}$ (°) | $\alpha$(°) |
| $S_1$ | 1.257 | 0.737 | 20.04 | 0.53 | 5.65 | 285,007 | 62.23 | 17.2 | 0.94 | 27.48 | 1386,164 | 46.70 | 11.6 |
| $S_2$ | 1.648 | 0.697 | 11.98 | 0.69 | 6.08 | 289,806 | 55.38 | 18.4 | 1.24 | 31.79 | 1516,477 | 38.98 | 12.0 |
| $S_3$ | 2.343 | 0.627 | 4.72 | 0.98 | 7.01 | 300,611 | 45.54 | 15.8 | 1.76 | 40.43 | 1734,770 | 29.65 | 9.93 |
| $S_4$ | 3.185 | 0.542 | 1.12 | 1.33 | 8.34 | 309,220 | 36.85 | 10.7 | 2.39 | 51.78 | 1920,642 | 22.72 | 6.61 |
| $S_5$ | 5.532 | 0.305 | -2.50 | 2.32 | 12.62 | 263,396 | 23.34 | 0.84 | 4.15 | 85.33 | 1781,168 | 13.55 | 1.05 |

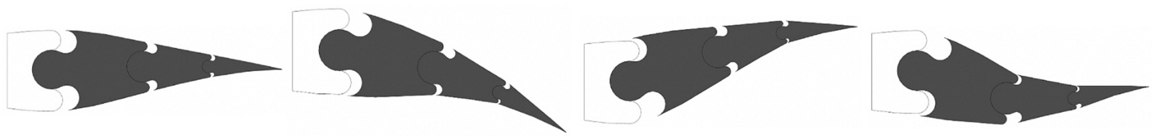

**Fig. 2.** Wind turbine blade morphing concept: Morphable, finger-like, multi-elements trailing edge.

rotating independently from each other, either upwards or downwards (see Fig. 2). The practical implementation of such a morphable blade is beyond the scope of the present study. However, a preliminary design was explored from a conceptual perspective, and is currently investigated within another specific research framework (Fong, 2024). This will be the matter of a subsequent publication (Fong et al., 2024).

## 3. Computational methodology (including validation)

This section presents the computational approach and set-up, which are validated against computational and experimental data coming from the literature (Wolfe and Ochs, 1997).

### 3.1. Computational matrix

As said above, the high versatility of the present morphing concept relies on the numerous combinations that can be obtained by deflecting each one of the three trailing edge elements independently of the others. Since exploring all possible options would deem infeasible from a computational viewpoint, here, only a few selected morphed configurations are considered. They correspond to an airfoil whose downstream elements would be homogeneously deflected (either downward or upward) by an identical angle ($\theta$) corresponding to one third of the overall cumulative deflection value ($\theta_{morph} = 3\theta$), cf. Fig. 3. A total of five configurations are explored, with $\theta_{morph}$ ranging from $-5°$ (morphed up) to $+15°$ (morphed down) by increments of 5°. Note that one of these configurations corresponds to the baseline airfoil, which is left unmorphed ($\theta_{morph} = 0°$).

For each of these (un)morphed configurations, a total of 32 flow conditions are considered, to cover situations that are representative enough (cf. Table 1). More precisely, four values of flow speed are chosen, corresponding to local Reynolds numbers $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$. For each flow regime ($Re$), a total of eight incidences are explored, with an angle-of-attack ($\alpha$) ranging from $\alpha_{\min} = -2.5°$ to $\alpha_{\max} = 15°$ by increments of 2.5°.

Overall, given the number of airfoil geometries (5), flow velocities (4) and incidences (8) considered, no less than 160 configurations are simulated, using an identical computational approach and set-up, which are introduced in the next section.

### 3.2. Computational approach and set-up

The numerical approach classically relies on Computational Fluid Dynamics (CFD), namely, two-dimensional (2D), steady, compressible simulations, which proved capable of accurately capturing the flow features around wing or blade sections. These simulations are performed using a well-known commercial CFD solver (ANSYS Fluent), which solves the Reynolds-Averaged Navier-Stokes (RANS) equations through a finite volume approach, using numerical schemes of 2nd order accuracy in both time and space (upwind schemes). The solver is used along with its 4-equation transition shear stress transport (SST) turbulence model, which advantageously mixes both a classical κ-ω turbulence model and its κ-ε counterpart. Indeed, whereas the former is known to be superior for resolving flows near the walls, the latter proved to be better suited for capturing flows away from the walls. Switching from one to the other model is achieved via a blending function, which is driven by the distance to the wall, thereby allowing for cross-diffusion when away from the solid boundaries but not near to them.

All simulations are conducted using 2D, C-type, structured meshes involving quadrilateral elements, which are known to be more accurate when employed with a finite volume method – as in here. The mesh involves cells of reasonably good quality, with acceptable average values in orthogonality (~ 0.996), skewness (~ $4\times10^{-3}$) and aspect ratio (~ 15) – including in the near-field region (cf. Fig. 4). The boundary layer region is made dense enough, with a first cell height of $1.24\times10^{-5}\,c$ (guaranteeing an $y+$ of 1.0 at the highest flow velocity, $Re = 2\times10^6$) along with a cells' growth rate of 1.15 in the radial direction. The azimuthal discretization is set to a total of 400 cells, upon a prior grid convergence which consisted in steadily increasing the number of wall points around the airfoil from 50 to 600 (cf. §3.3 and Fig. 5). Ultimately, the mesh comprises about 51,000 cells whilst extending over 15, 15 and 25 airfoil chords in the upstream, lateral and downstream direction, respectively. Fig. 4 depicts the resulting grid associated with the baseline airfoil, its morphed counterparts being derived in a similar fashion. Notably, upon a proper convergence study in terms of temporal discretization, a Courant Friedrich Lewy (CFL) number of 5 is retained for all simulations.

### 3.3. Validation of the computational methodology

The computational approach and set-up are first validated by using a reference case coming from the literature (Wolfe and Ochs, 1997), namely a classical S809 airfoil within a flow of Reynolds number $Re = 2.5\times10^6$ and variable incidences ($\alpha = 0°$ to $\alpha = 20°$). This configuration is here simulated using the unmorphed profile, which is allotted a unitary chord length ($c = 1$ m) and is exposed to an incoming airflow of velocity $v_\infty = 29.2$ m/s ($\rho = 1.225$ kg/m$^3$, $\mu = 1.79\times10^{-5}$ kg/m·s). Following (Wolfe and Ochs, 1997), a total of six incidences are computed ($\alpha = 0°$, 1°, 5°, 9.2°, 14.2°, 20.1°), the cases of lowest and highest incidence being also used for conducting the grid convergence

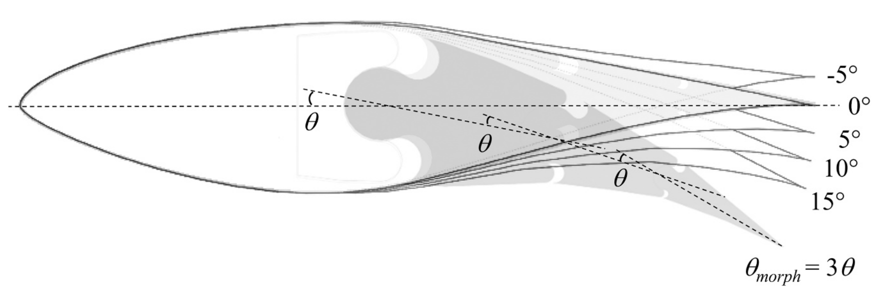


**Fig. 3.** Morphed configurations: NREL Phase VI wind turbine blade's generic section (S809 airfoil), when applied a morphing of variable extent (labelled via the cumulative deflection angle, $\theta_{morph} = 3\theta$).

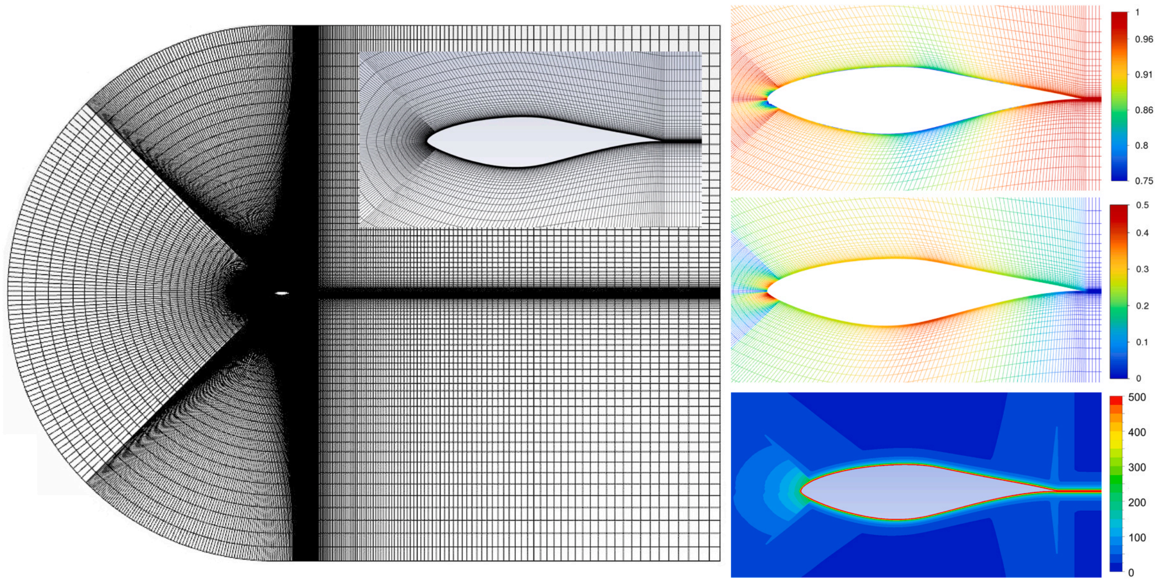


**Fig. 4.** Computational mesh (left) and corresponding quality criteria (near-field cells' orthogonality, skewness and aspect ratio – right, from top to bootom, respectively).

mentioned above (cf. Fig. 5). All calculation results are compared to both the experimental and computational data reported in Wolfe and Ochs, 1997. As can be seen in Fig. 6, the agreement is good, as proven by the close values obtained in terms of lift and drag coefficients. More precisely, the present simulations match very closely the experimental data, at least for what concerns the low-to-moderate flow incidences. At higher incidences ($\alpha > 7.5°$), the agreement is a bit less satisfactory, especially for what concerns the drag coefficient. This is likely to result from the stall onset occurring beyond this angle-of-attack, which makes it more challenging for the CFD to capture accurately the transition point. Despite these slight discrepancies at higher incidences, the agreement is very favorable, overall. Notably, the present simulation matches the experiment better than the reference calculation, which relied on a tailored transition SST model and failed to capture the lift at moderate-to-high incidences ($\alpha > 10°$) (Wolfe and Ochs, 1997).

Upon the above validation, the computational approach and set-up is kept identical for the rest of the study, the S809 profile and 2D mesh being each time adjusted so as to reproduce the desired morphing.

## 4. Computational results and analysis

As said above, the computational study consisted in exploring the aerodynamics of a S809 profile with five degrees of morphing applied ($\theta_{morph} = -5°$ to $+15°$), when exposed to incoming flows of four Rey-

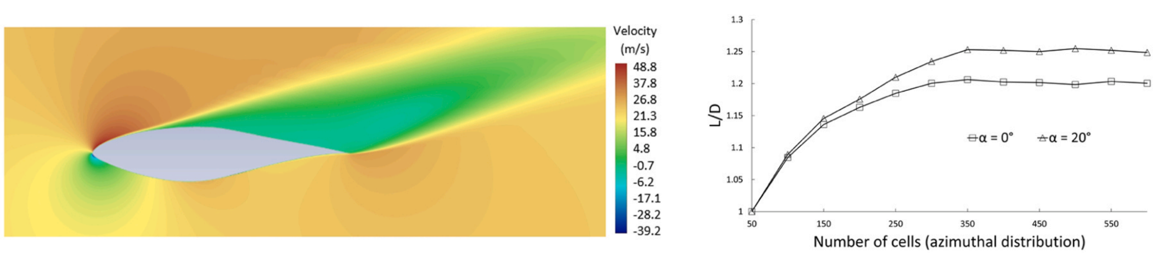


**Fig. 5.** Validation of the computational approach via steady CFD simulations of the unmorphed blade's generic section (S809) within a $Re = 2.5 \times 10^6$ flow at various incidences. Left: Snapshot of the axial velocity ($\alpha = 20°$). Right: Grid convergence study (lift-to-drag ratio at lowest and highest incidences i.e. $\alpha = 0°$, $20°$).

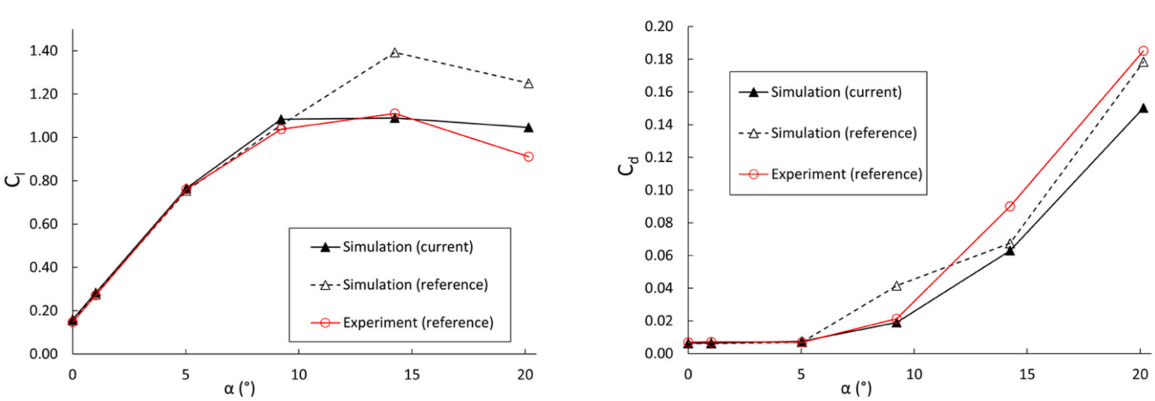


**Fig. 6.** Validation of the computational approach via steady CFD simulations of the unmorphed generic blade section (S809) within a $Re = 2.5 \times 10^6$ flow at various incidences. Polar in lift (left) and drag (right) coefficients, with comparison against reference computational and experimental results (Wolfe and Ochs, 1997).

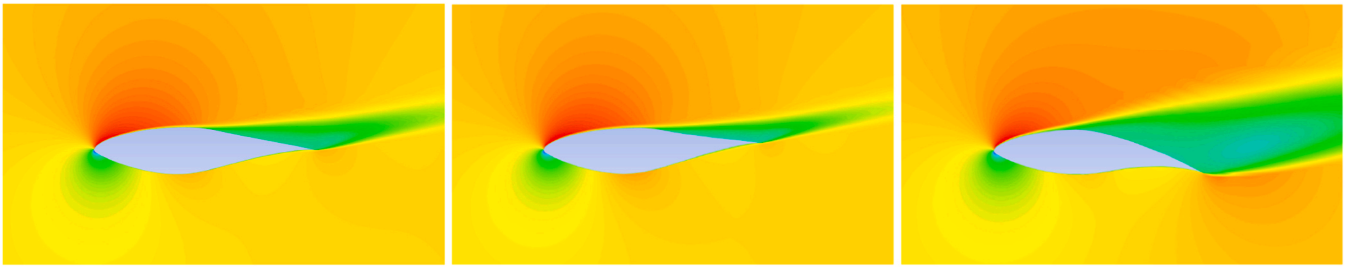

**Fig. 7.** Steady mean field (axial velocity) characterizing the unmorphed blade section (left) and two of its morphed counterparts ($\theta_{morph} = -5°$ and $15°$, center and right), as simulated for the flow of highest speed ($Re = 2.0{\times}10^6$) and incidence ($\alpha = 15°$).

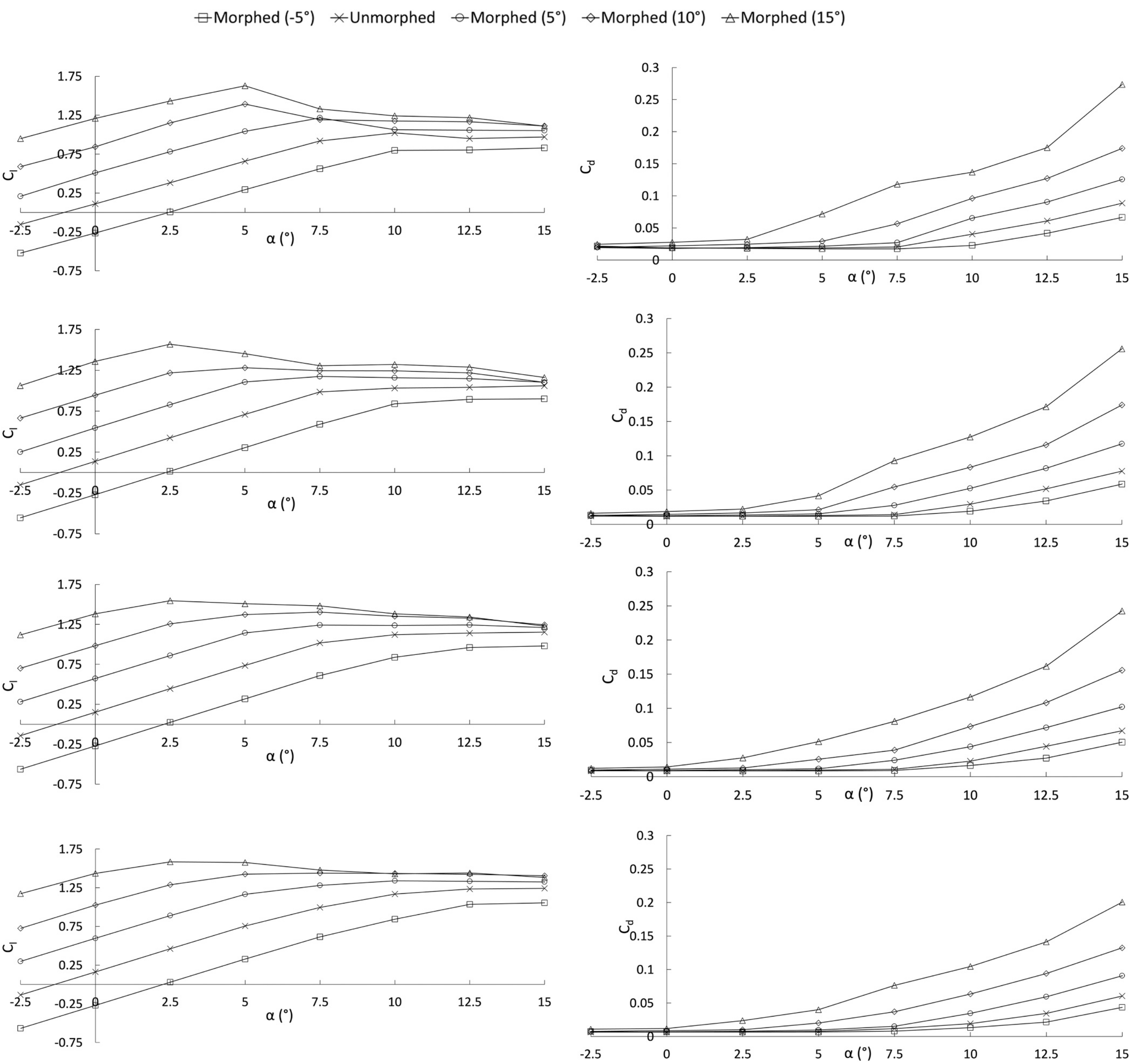


**Fig. 8.** Aerodynamic characteristics exhibited by the blade section ($S_{1\text{–}5}$) when morphed ($\theta_{morph} = -5°$ to $15°$) within flows of Reynolds number $Re = 0.25{\times}10^6$, $0.5{\times}10^6$, $1{\times}10^6$ and $2{\times}10^6$ (from top to bottom). Polars in lift (left) and drag (right) coefficients.

nolds numbers ($Re = 0.25{\times}10^6$, $0.5{\times}10^6$, $1{\times}10^6$, $2{\times}10^6$) and eight incidences ($\alpha = -2.5°$ to $15°$). The present section summarizes the computational results obtained across these 160 configurations, 3 of which are exemplified in Fig. 7; This figure depicts the steady mean field in axial velocity obtained for the baseline and two morphed ($\theta_{morph} = -2.5°$, $15°$) airfoils, when exposed to the flow of highest speed ($Re = 2{\times}10^6$) and incidence ($\alpha = 15°$).

### 4.1. Aerodynamic performances by all airfoils

Fig. 8 depicts the lift and drag polars associated with all five airfoils, as obtained for each Reynolds number ($Re = 0.25\times10^6$ to $2\times10^6$, from top to bottom). As can be seen, compared to the baseline (unmorphed) one, the positively (resp. negatively) morphed configurations exhibit a higher (resp. lower) lift. This phenomenon is proportional to the extent of morphing ($\theta_{morph}$), but with a proportionality factor that depends on the flow condition (incidence and/or speed). Consequently, the lift gain can be high, albeit very variable. For instance, when positively morphed (i.e., with its trailing edge deflected downward), the airfoil exhibits a lift that varies from 1 to 8.5 times that of its unmorphed counterpart, depending on the conditions (morphing level, flow speed and incidence). Overall, it appears that, whatever the flow speed, the relative gain in lift offered by higher levels of morphing gets narrower as the flow incidence increases. Alternatively, the highest morphing ($\theta_{morph} = 15°$) systematically maximizes the lift across all incidences ($\alpha$ = -2.5° to 15°) at low-to-moderate flow speeds ($Re = 0.25\times10^6$, $0.5\times10^6$), whereas being challenged by a lesser morphing ($\theta_{morph} = 10°$) at higher flow speeds ($Re = 1\times10^6$, $2\times10^6$) and incidences ($\alpha \geq 10°$). In other words, despite a clear proportionality between the extent of morphing applied and the lift gain observed, the relationship is not linear, and a saturation effect occurs once certain levels of flow incidence or speed are reached. Upon what precedes, it appears that enhancing (resp. diminishing) the lift generation by a blade can be obtained by positively (resp. negatively) morphing its sections, even though the gain is seen to depend on the flow conditions (speed and incidence), with a plateau reached when the latter are too high.

Even though the lift is the main driver for the aerodynamic efficiency – and structural integrity – of wind turbine blades, it is not the sole one since drag also plays a role (cf. Eq. (3)). As shown on right side of Fig. 8, whatever the airfoil and flow conditions, the lift changes resulting from the morphing effect unavoidably incur a variation in the drag. The latter appears to become more and more prominent as both the morphing level and flow incidence increase, whereas continuously diminishing as the flow speed gets higher. Interestingly enough, these trends are different than those observed on the lift variations, thus making the analysis of morphing benefits less straightforward.

Therefore, to make things more tractable, Fig. 9 depicts the lift-to-drag ratio polars associated with each flow regime ($Re$). From there, several observations can be made; First, whether the airfoil is morphed or not, its $L/D$ ratio continuously increases as the flow speed gets higher, being roughly magnified by a factor 2.5 as the Reynolds number ranges from $Re = 0.25\times10^6$ to $Re = 2\times10^6$. Second, each airfoil is characterized by a specific optimal incidence (namely, the stall angle-of-attack), which maximizes its $L/D$ ratio while being (roughly) independent of the Reynolds number. For all positively morphed airfoils ($\theta_{morph} = 5°$, $10°$ and $15°$), this optimal incidence is inversely proportional to the extent of morphing ($\alpha_{opt} \approx 5°$, $2.5°$ and $0°$, respectively), the resulting maximal $L/D$ ratio being nevertheless roughly identical (e.g., $L/D_{max} \approx 45$, 70, 100 and 120 for $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$, respectively). Notably, this maximum $L/D$ ratio value is similar to that of the unmorphed airfoil, which is however reached at a higher incidence ($\alpha_{opt} \approx 7.5°$) – except when the flow speed is the highest $Re = 2\times10^6$), in which case it is less and occurs at a lesser incidence ($\alpha_{opt} \approx 5°$). On another hand, the negative morphing airfoil ($\theta_{morph} = -5°$) exhibits a $L/D$ ratio that is systematically lower than that of its unmorphed counterpart, except when the flow incidence is beyond its optimal value ($\alpha_{opt} \approx 7.5°$ to $10°$, depending on the Reynolds).

At this stage, it should be noted that the above values of optimal incidence ($\alpha_{opt}$) are approximative only, given the rather large increments – and thus, error margins – in angle-of-attack ($\Delta\alpha = 2.5°$). Refining this point would imply covering more flow conditions, which is deemed difficult considering the already consequent number of simulations conducted (160+). Nevertheless, despite these uncertainties, the above lift-to-drag polars exhibit a clear trend; Whatever the flow regime ($Re$), the $L/D$ ratio exhibited by the positively (resp. negatively) morphed airfoils is higher than that of their baseline (unmorphed) counterpart whenever the flow incidence is below (resp. above) the latter's stall angle ($\alpha_{opt}$).

Finally, as said above, the optimal metric for quantifying the performances of a wind turbine blade is its aerodynamic power coefficient. The latter is proportional to the lift coefficient, with a proportionality factor that is driven by the lift-to-drag ratio ($C_L/C_D$) and the advancement-to-wind speeds ratio ($\lambda$) – see Eq. (3). Therefore, to further discriminate the aerodynamic merits of all airfoils, Fig. 10 depicts their respective power coefficient polar, as obtained for each flow regime ($Re$). This is here achieved for the particular case of blade section $S_4$, for which the considered local Reynolds values ($Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$) would correspond to wind speeds $v_\infty \cong 5$, 8, 15 and 20 m/s, along with blade rotational rates $\omega \cong 13.5$, 32.5, 67.5 and 150 RPM, thereby leading to advancement-to-wind speeds ratio $\lambda \cong 0.9$,

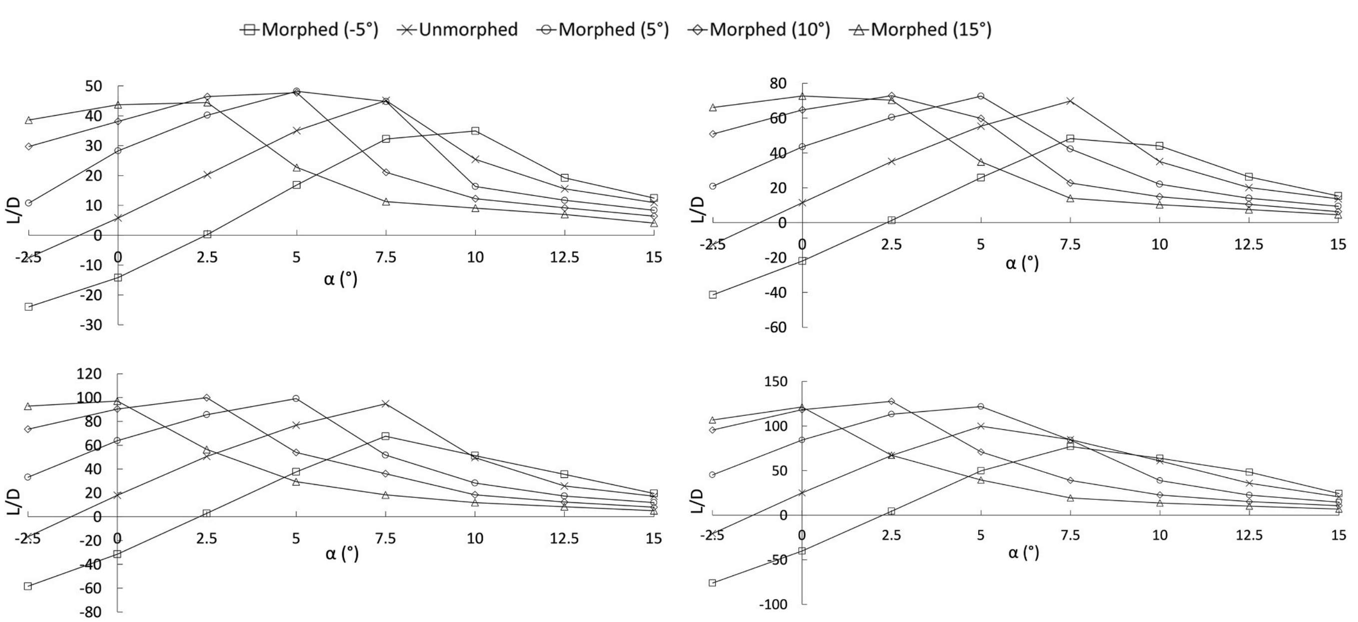


**Fig. 9.** Aerodynamic characteristics exhibited by any blade section ($S_{1-5}$) when morphed ($\theta_{morph} = -5°$ to $15°$) within flows of Reynolds number $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$ (from left to right and top to bottom). Polars in lift-to-drag ratio.

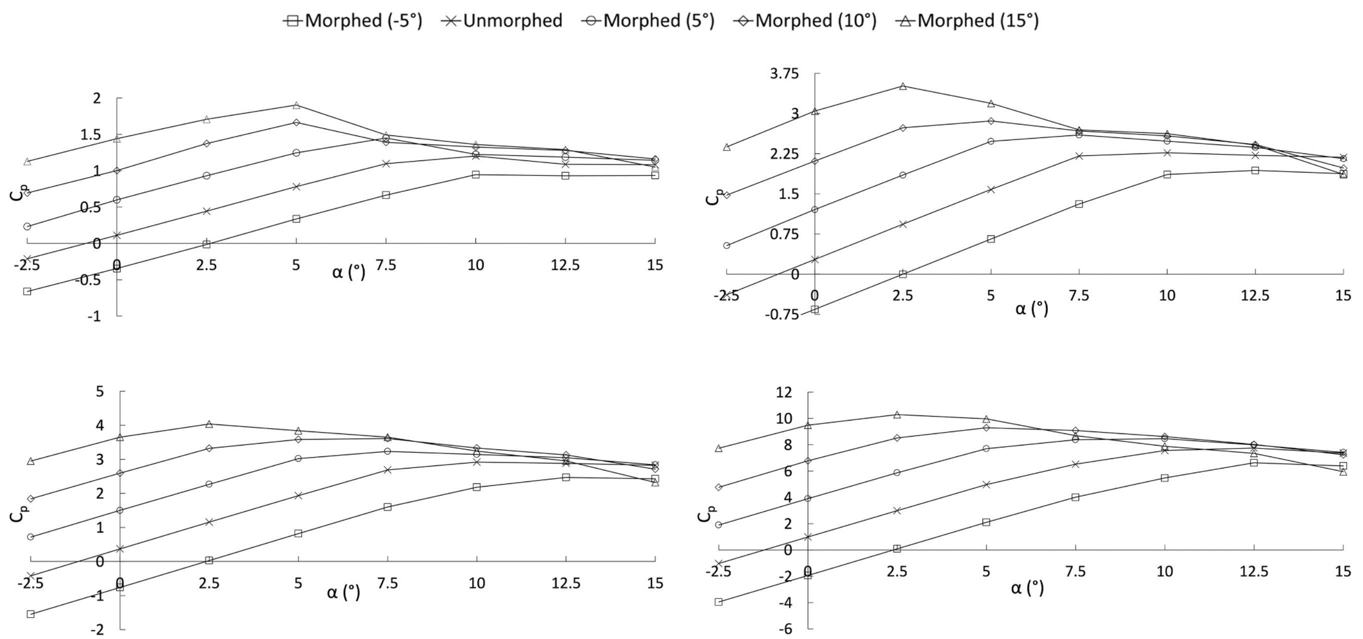


**Fig. 10.** Aerodynamic power characteristics exhibited by the blade section $S_4$ when morphed ($\theta_{morph}$ = −5° to 15°) within flows of Reynolds number $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$ (from left to right and top to bottom). Polars in power coefficient.

1.35, 1.5 and 2.5 (with a variable incidence $\alpha$, depending on the blade pitch $\beta$ applied).

By comparing these $C_P$ polars with their $C_L$ counterparts (Fig. 8, left side), one can notice that the trend is similar, which naturally reflects the proportionality between power and lift coefficients. These trends, however, are not identical, especially for what concerns the higher values of flow incidence ($\alpha > 5°$) – which translates the more detrimental effect of the drag. Despite this, the gain in performances brought by the morphing is significant, with a power coefficient that may increase by more than one order of magnitude in some situations. As an example, for the flow of lowest Reynolds ($Re = 0.25\times10^6$) under no incidence ($\alpha = 0°$), the positively morphed airfoils ($\theta_{morph}$ = 5°, 10° and 15°) exhibit a power coefficient that is respectively 5.3, 8.9 and 12.8 times higher than that of their unmorphed counterpart. These $C_P$ gains are seen to diminish as the flow incidence ($\alpha$) – and, to a lesser extent, speed ($Re$) – increase, very likely because of the higher drag incurred. They, however, remain positive across all flow speeds and incidences, with – each time – at least one morphed configuration delivering more (or as much) power as its baseline (unmorphed) counterpart.

Upon all the above, it is deemed possible to optimize the aero-structural behavior of a wind turbine blade under any given wind force and rotational speed, by simply morphing each one of its sections in a *tailored* fashion, to adjust its aerodynamic response (lift coefficient, lift-to-drag ratio) to the local flow conditions (flow speed or incidence). Illustrating this point is the matter of the next subsection.

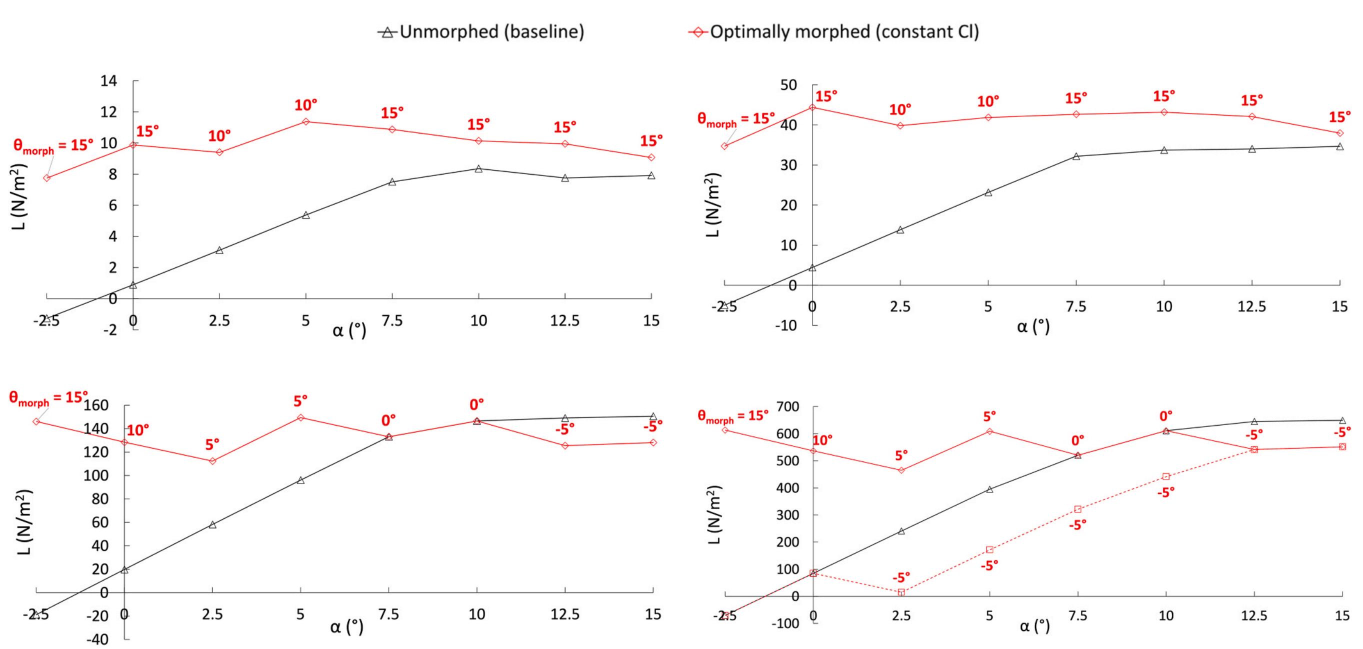


**Fig. 11.** Morphing strategies for alleviating the aero-structural loads experienced by the blade: Lift exhibited by any blade section $S_{1\text{-}5}$ when optimally morphed for roughly constant (red line) or minimal (red dashes) aerodynamic loads, within flows of various Reynolds numbers ($Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$, from left to right and top to bottom). Polars in lift force, with comparison against that of the unmorphed blade section (black line).

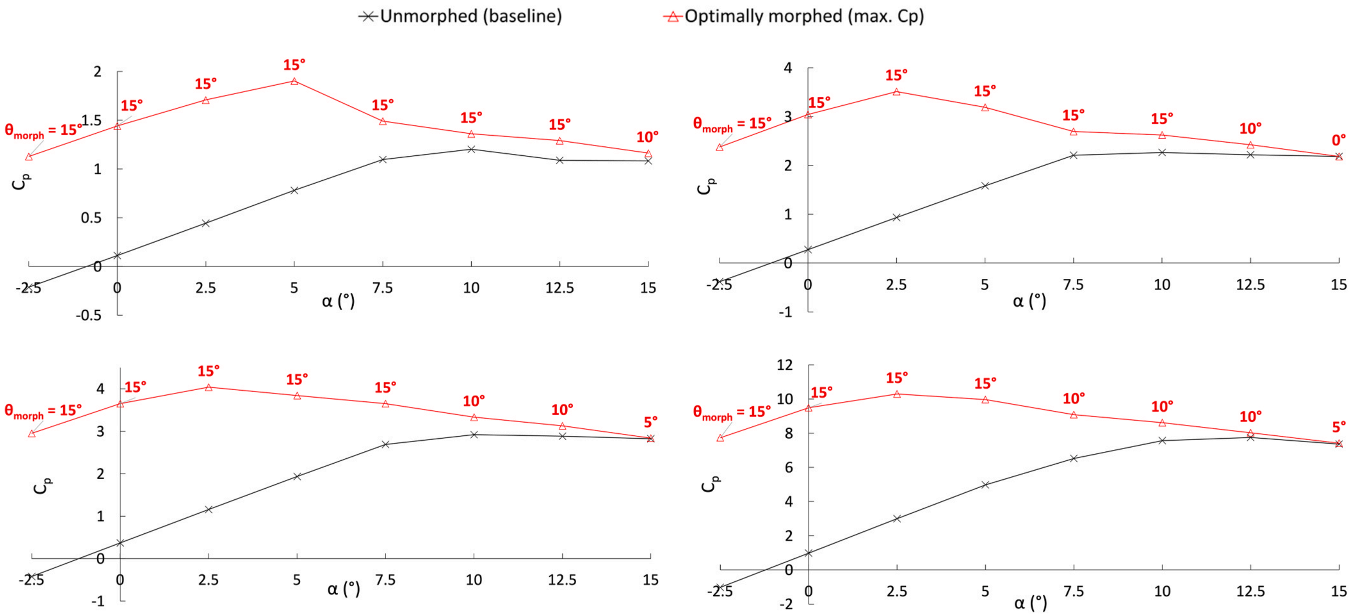


**Fig. 12.** Morphing strategies for maximizing the power generated by the blade: Aerodynamic power exhibited by the blade section $S_4$ when optimally morphed for a maximum power coefficient (red line), within flows of various Reynolds numbers ($Re = 0.25{\times}10^6$, $0.5{\times}10^6$, $1{\times}10^6$ and $2{\times}10^6$, from left to right and top to bottom). Polars in power coefficient, with comparison against that of the unmorphed blade section (black line).

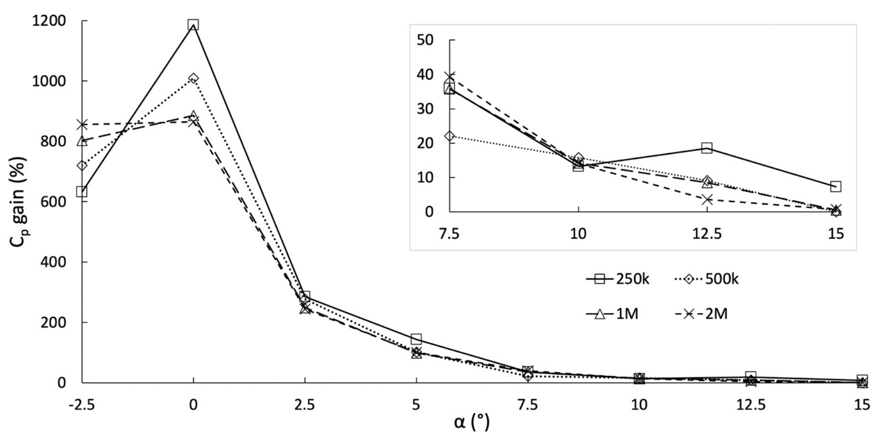


**Fig. 13.** Morphing strategies for maximizing the power generated by the blade: Gain in the aerodynamic power exhibited by the blade section $S_4$ when optimally morphed for a maximum power coefficient within flows of various Reynolds numbers ($Re = 0.25{\times}10^6$, $0.5{\times}10^6$, $1{\times}10^6$ and $2{\times}10^6$, labelled as“250k”, “500k”, "1M" and "2M", respectively). Polars of the gain in power coefficient, with respect to the unmorphed blade section.

### 4.2. Optimal morphing scenarios

Upon what precedes, one can infer various scenarios for an optimal morphing, depending on what would be the operational conditions (flow speed or incidence) and objectives (e.g., maximizing the aerodynamic performance vs. alleviating the structural loads experienced by the blade section).

For instance, in case of excessive aerodynamic loading due to high-speed winds, the concerned blade sections could be morphed negatively, to alleviate the aerodynamic loads that would be produced otherwise. Indeed, as was shown in bottom/left of Fig. 8, whatever the flow speed or incidence, the negative morphing ($\theta_{morph} = -5°$) incurs a lift that is systematically lower than that of its unmorphed counterpart, with a lift alleviation that can go up to 94%. More important, across all flow speeds (including the highest one, $Re = 2{\times}10^6$), the negatively morphed airfoil generates a lift that is *nil* or *negative* whenever the flow incidence is low enough ($\alpha = -2.5°$, $0°$, $2.5°$).

Oppositely, instead of minimizing the lift produced, one could seek to rather maintain it to a nominal and *stable* value across all operational conditions, this being done by applying the blade section a positive

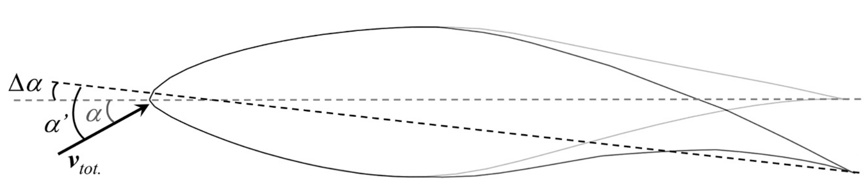


**Fig. 14.** Further exploration of the morphing effect: Deflection of the airfoil’s chordline incurred by the morphing, with subsequent alteration of the angle-of-attack ($\Delta\alpha$).

**Table 2**
Further exploration of the morphing effect: Variation of the flow incidence ($\Delta\alpha$) incurred by the trailing edge deflection resulting from the morphing ($\theta_{morph}$).

| $\theta_{morph}$ | -5° | 0° | 5° | 10° | 15° |
|---|---|---|---|---|---|
| $\Delta\alpha(\theta_{morph})$ | -2.02° | 0° | 1.96° | 3.97° | 5.93° |

morphing to be tailored to the flow incidence (i.e., blade rotational speed). As an illustration of the previous considerations, Fig. 11 compares the lift polar of the baseline (unmorphed) airfoil to that obtained for its optimally morphed counterparts; Being tailored to the flow speed ($Re$), the latter ensures a roughly constant lift across all flow incidences (red lines) or – oppositely – guarantees a minimal loading for the particular case of highest flow speed ($Re = 2\times10^6$, red/dashed lines in the bottom/right image).

Finally, one could rather seek to maximize the aerodynamic power through an alternative morphing strategy. This is illustrated in Fig. 12, which exemplifies how the power delivered by any given blade section (here, $S_4$) could importantly benefit from a morphing tailored to the local flow condition (speed and incidence). Compared to its baseline (unmorphed) counterpart, the optimally morphed airfoil exhibits a power coefficient that is both magnified (sometimes significantly) and made almost insensitive to the flow incidence. This would translate into an increased and more stable power delivery by the rotor, across all rotational speeds. Notably, this important and relatively constant power coefficient characterizing the optimally morphed blade section stems from the morphing strategy itself; The latter, which is rather robust – albeit not identical – across all $Re$ regimes, consists in exerting the airfoil a morphing degree that is inversely proportional to the flow incidence, decreasing regularly as the values in angle-of-attack gets higher. In a way, the morphing acts as a means to supplement the lack of aerodynamic performances exhibited by the baseline airfoil over the lower incidences, whatever the flow speed. Ultimately, these $C_P$ polars characterizing the optimally morphed blade section translate into an important gain in power coefficient over the low-to-moderate flow incidences. This is made more explicit in Fig. 13, which depicts the gain in power coefficient resulting from the present $C_P$-optimal morphing strategy. As can be seen, the performance increase is rather spectacular, with $C_P$ gains that can go as high as 1200%, e.g., for the slowest flow at no incidence ($Re = 0.25\times10^6$, $\alpha = 0°$). These benefits diminish slightly as the flow speed increases (e.g., a gain of 800% at $Re = 2\times10^6$ and $\alpha = 0°$), whereas degrading more importantly when the incidence goes up. Still, the gain remains substantial for non-zero incidences across all flow regimes, namely about 100%, 30–40% and 10–15% when $\alpha = 5°$, $\alpha = 7.5°$ and $\alpha = 15°$, respectively.

### 4.3. Further exploration of the morphing effect

At this stage, it is worth exploring the phenomenology behind the aerodynamic benefits offered by the morphing. For instance, one could legitimately wonder if these gains do not simply originate from the chord line deflection – rather than the shape deformation – that the morphing incurs onto the airfoil. Indeed, when morphed, the airfoil sees its trailing edge deflected ($\theta_{morph} = 3\theta$), which unavoidably entails a variation of its angle-of-attack ($\Delta\alpha$) since the latter is classically defined with respect to the chord line connecting its leading and trailing edges (see Fig. 14 and Table 2). Because of this chord line deflection, a given morphed airfoil ($\theta_{morph}$) set at an *actual* incidence $\alpha$ is thus to exhibit an *apparent* one, $\alpha' = \alpha + \Delta\alpha(\theta_{morph})$. One could then argue that it would suffice to set the *unmorphed* airfoil at this apparent incidence ($\alpha'$) for it offers the same aerodynamics merits than its morphed counterpart under the actual incidence ($\alpha$). In which case, the blades' aerodynamic performances could be enhanced more simply, through the application of a distributed *local* pitching (i.e., a global twisting), instead of a morphing strategy.

This point is here explored by characterizing the aerodynamics of the unmorphed airfoil, when set at apparent incidences ($\alpha'$) corresponding to the actual ones ($\alpha$) of its morphed counterparts. For the sake of simplicity, such alternative characterization of the unmorphed airfoil for new, apparent incidences is here made through a simple linear interpolation of its baseline results in lift and drag coefficients (Fig. 8, unmorphed airfoil curves). Please, note however that the validity of this analytical manipulation was verified through a direct comparison against the *exact* results coming from a few additional simulations, which specifically computed the unmorphed airfoil under the corresponding apparent incidences.

With respect to the optimal morphing strategy maximizing the aerodynamic power, Fig. 15 depicts the $C_P$ polars inferred for the unmorphed airfoil, when set at apparent incidences corresponding to the actual ones initially covered by both the unmorphed and morphed

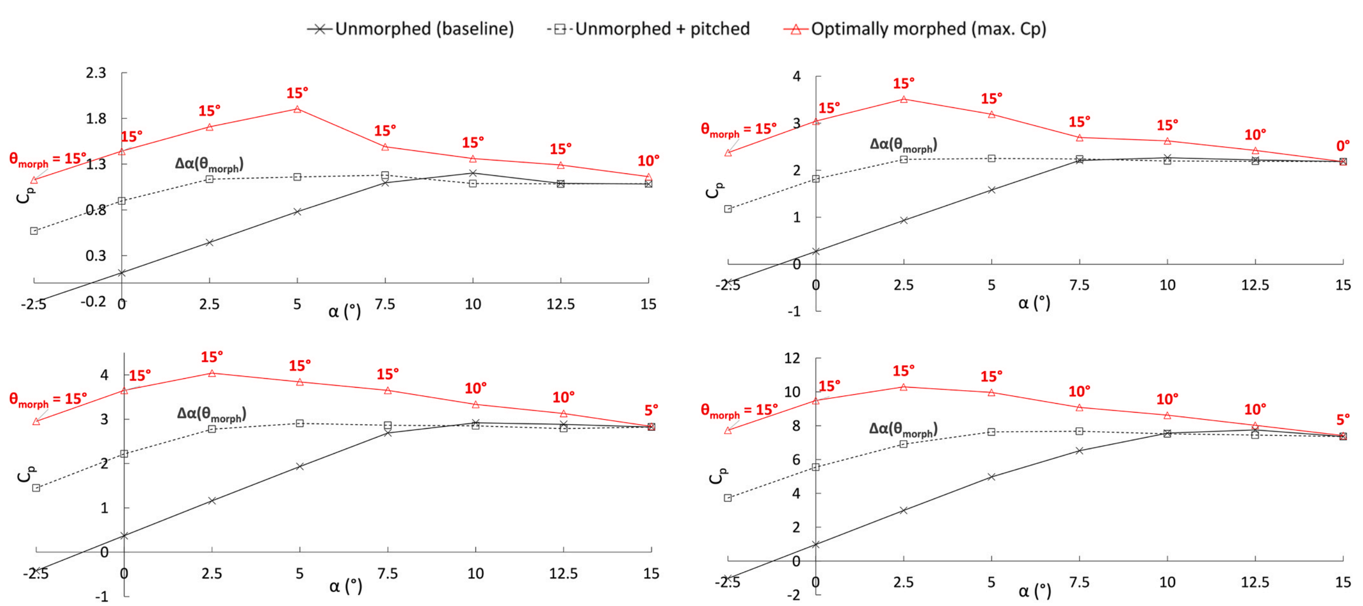


**Fig. 15.** Further exploration of the morphing effect: Aerodynamic power characteristics exhibited by the blade section $S_4$ when optimally morphed (red line) or solely pitched (black dashes) within flows of Reynolds number $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$ (from left to right and top to bottom). Polars in power coefficient, with comparison against that of the unmorphed / unpitched blade section (black line).

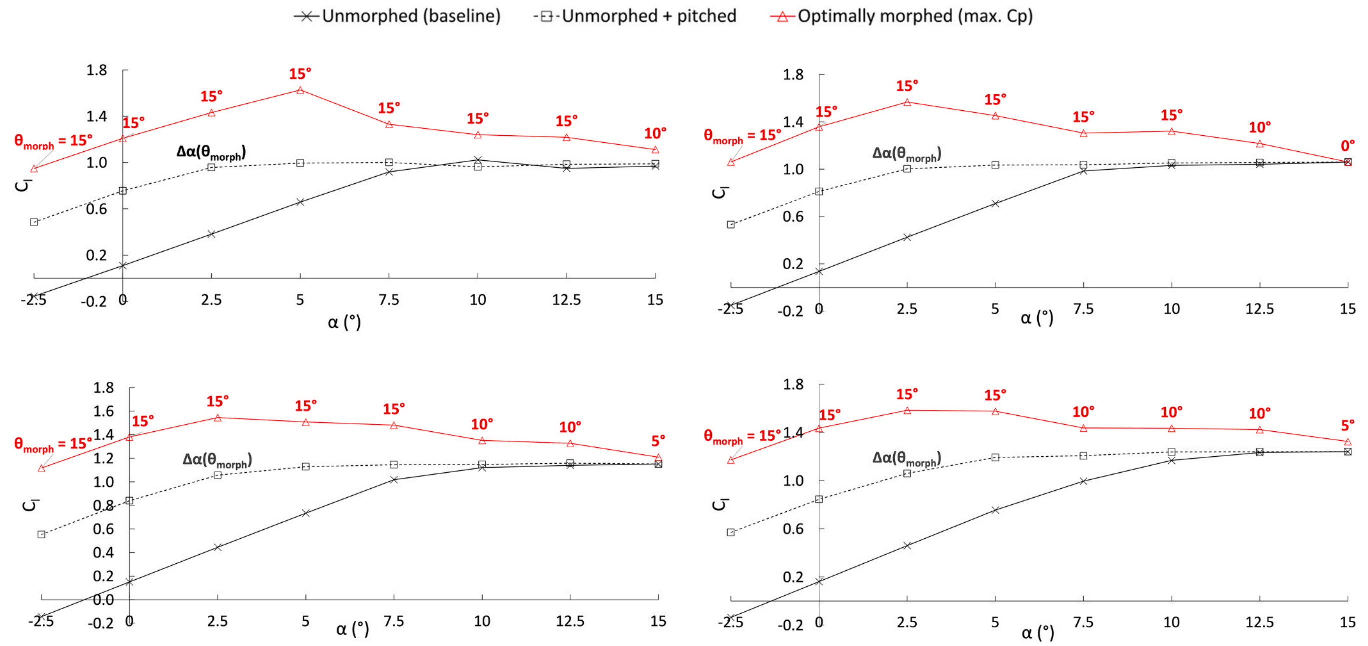


**Fig. 16.** Further exploration of the morphing effect: Aerodynamic characteristics exhibited by the blade section $S_4$ when optimally morphed (red line) or solely pitched (black dashes) within flows of Reynolds number $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$ (from left to right and top to bottom). Polars in lift coefficient, with comparison against that of the unmorphed / unpitched blade section (black line).

airfoils (whose results are reproduced here, for comparison). More precisely, whereas the black and red lines still depict the results obtained for the unmorphed and morphed airfoils (cf. Fig. 12), the black dash curves now exhibit the outputs coming from the present alternative configuration of an unmorphed but *pitched* airfoil. As can be seen, pitching the (unmorphed) airfoil contributes to less than half of the power gain brought by its morphing, overall. Since this pitching effect is deemed to be representative of the morphed airfoil's chord line deflection, the remaining $C_P$ gain can be reasonably attributed to its actual shape deformation. It is worth noticing that this observation stands across all flow speeds and incidences. Notably, at mid-to-high incidences ($\alpha \geq 7.5°$), the pitching effect is almost nil, i.e., the (modest) $C_P$ gain brought by the morphing is *solely* due to its shape deformation rather than its chord line deflection.

The corresponding polars in lift and drag coefficients are respectively depicted in Fig. 16 and Fig. 17, which compare the results from all three airfoil configurations (unmorphed, morphed, unmorphed but pitched). As revealed by Fig. 16, the trend observed for the lift coefficient resembles that of its power counterpart; Whatever the flow speed ($Re$), the chord line deflection effect accounts for roughly half of the gain observed over low-

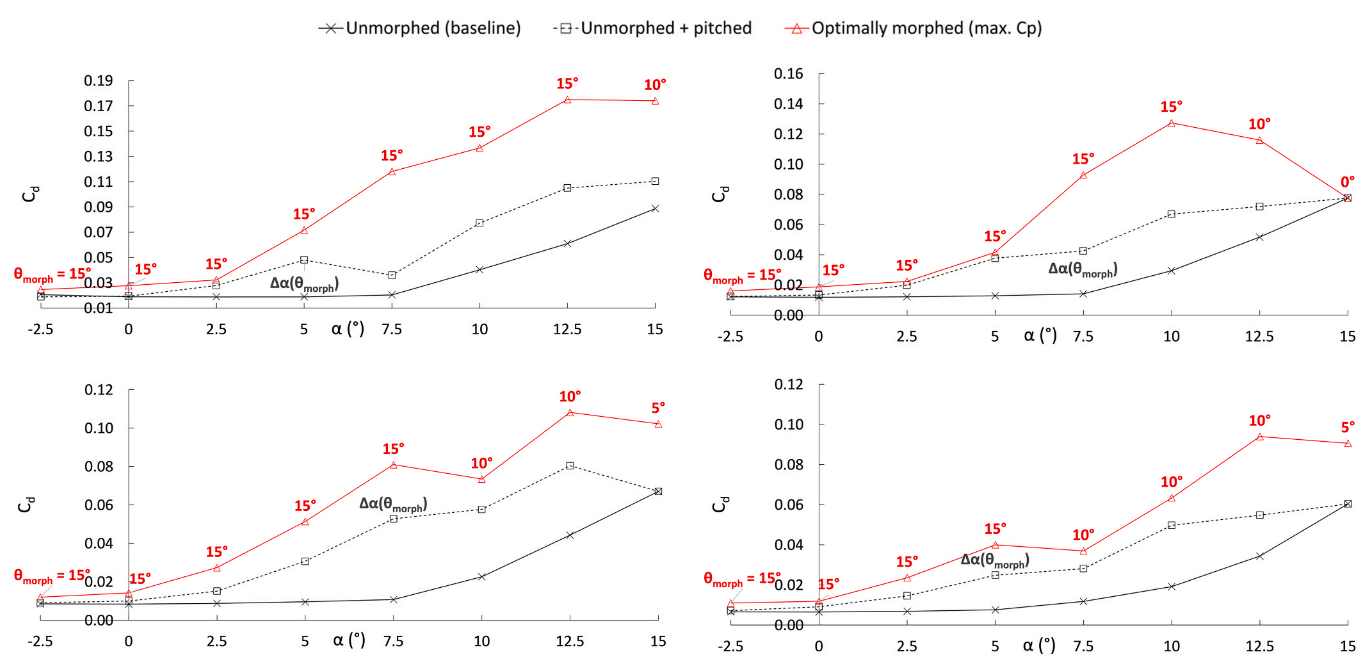


**Fig. 17.** Further exploration of the morphing effect: Aerodynamic characteristics exhibited by the blade section $S_4$ when optimally morphed (red line) or solely pitched (black dashes) within flows of Reynolds number $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$(from left to right and top to bottom). Polars in drag coefficient, with comparison against that of the unmorphed / unpitched blade section (black line).

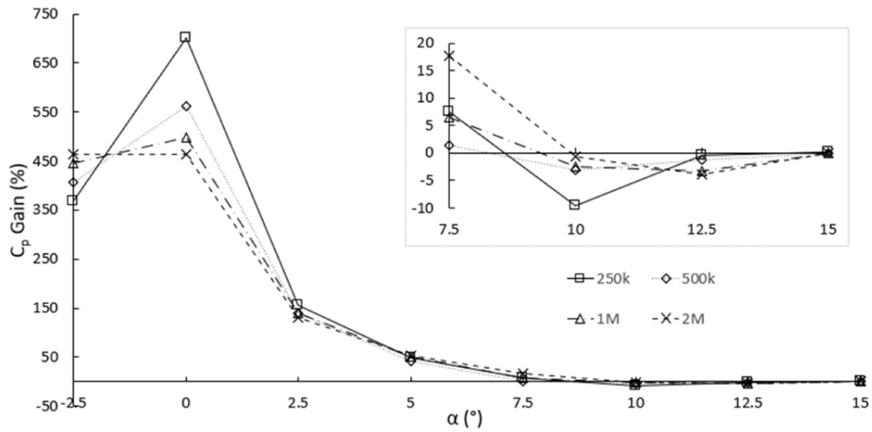


**Fig. 18.** Further exploration of the morphing effect: Aerodynamic power characteristics exhibited by the blade section $S_4$ when unmorphed but pitched within flows of various Reynolds numbers $Re = 0.25\times10^6$, $0.5\times10^6$, $1\times10^6$ and $2\times10^6$, labelled as "250k", "500k", "1M" and "2M", respectively. Polars of the gain in power coefficient, with respect to the unmorphed / unpitched blade section.

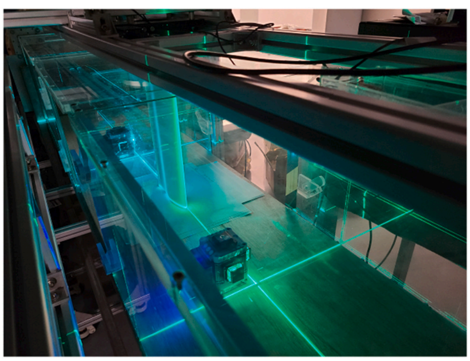


**Fig. 19.** Experimental apparatus: Wind tunnel vein (left) and mock-ups of a morphed blade section (right).

to-mid incidences ($\alpha \leq 7.5°$), and for almost none of it over higher ones. Therefore, here too, the remaining part of the gain can reasonably be credited to the sole shape deformation brought by the morphing.

On another hand, Fig. 17 reveals that the chord line deflection effect accounts for a non-negligible fraction of the drag penalty, especially at lower incidences ($\alpha$), where the excess drag by the unmorphed but pitched airfoil can be as important as that of its morphed counterpart (black dashes and red lines, respectively). This trend, however, does not subsist at higher incidences, where the morphing incurs a more important drag than its unmorphed/pitched counterpart – which is likely to explain the saturation effect in the power gain previously observed at high angle-of-attack values.

The above results explain why the sole effect of pitching (or chord line deflection) contributes to only a fraction of the power benefits offered by the morphing. To make this observation more explicit, Fig. 18 depicts the $C_P$ gain resulting from the sole pitching effect, which can be compared to that of the morphing (cf. Fig. 13). A straight comparison of these two figures reveals that even though the trends are qualitatively similar, they differ importantly from a quantitative viewpoint; Overall, the pitching effect contributes to roughly half of the power gain brought by the morphing, across all flow speeds and incidences. Notably, this pitching effect turns out to be detrimental for higher flow incidence ($\alpha \geq 10°$), with a $C_P$ gain that then becomes *negative* – contrarily to its morphing counterpart (cf. Fig. 13).

## 5. Experimental validation

To validate the methodological and phenomenological conclusions drawn above, an experimental campaign is conducted via small-scale experiments whose set-up is detailed before the test results are discussed hereafter.

### 5.1. Experimental means and setup

The experiment is conducted in the Small Wind Tunnel (SWT) from the Aerodynamic and Acoustics Facility (AAF) of The Hong Kong University of Science and Technology (HKUST). The SWT facility is a closed-vein, hard-walled wind tunnel whose test section extends over 2.43 m × 0.5 m × 0.4 m in the axial, lateral and vertical directions, respectively (see Fig. 19). The vein undergoes a low-speed, incompressible flow whose maximum freestream velocity is $U_\infty = 24.5$ m/s, thereby corresponding to an upper Mach number $M = 0.074$. The turbulence intensity at the test section center is lower than 0.4%, thereby ensuring a flow that is clean enough.

Designing, manufacturing and operating a complete morphable airfoil mock-up would substantially increase the experimental complexity without bringing real added value in terms of steady flow characterization. Therefore, the experiment is rather based on a set of five, distinct, rigid airfoils, each reproducing one of the considered morphed configurations. More precisely, all five models correspond to a S809 bidimensional profile whereas differing solely by their trailing edge section, which is either applied a specific morphing ($\theta_{morph} = -5°$, $+5°$, $+10°$, $+15°$, see Fig. 19) or left untouched – thus corresponding to the unmorphed, baseline case, $\theta_{morph} = 0°$. All models are 3D-printed out of Somos GP resin through a Multi-Jet Fusion (MJF) printing process, thereby guaranteeing a high level of accuracy (smooth surface, with tight tolerances).

For each configuration tested, the corresponding airfoil model is installed within the vein, being mounted vertically on the flow centerline (see Fig. 19). This allows pitching the airfoil within the vein's lateral plane, whose higher dimensions are expected to mitigate the possible installation effects (*e.g.*, flow blockage, downwash). Still for minimizing these effects, the airfoil's pitch axis is located half-way through its

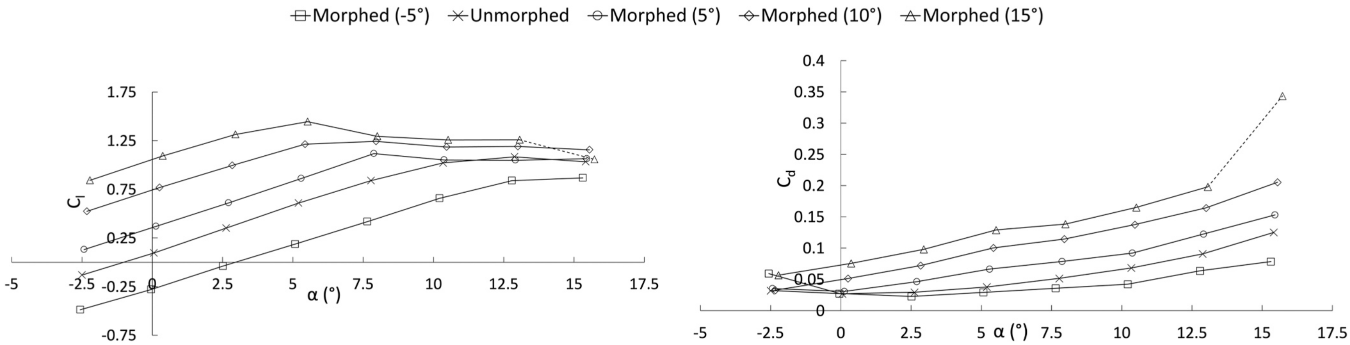


**Fig. 20.** Aerodynamic characteristics exhibited by the morphed blade section mock-ups, when tested within a flow of Reynolds number $Re = 0.25\times10^6$. Polars in lift (left) and drag (right) coefficients.

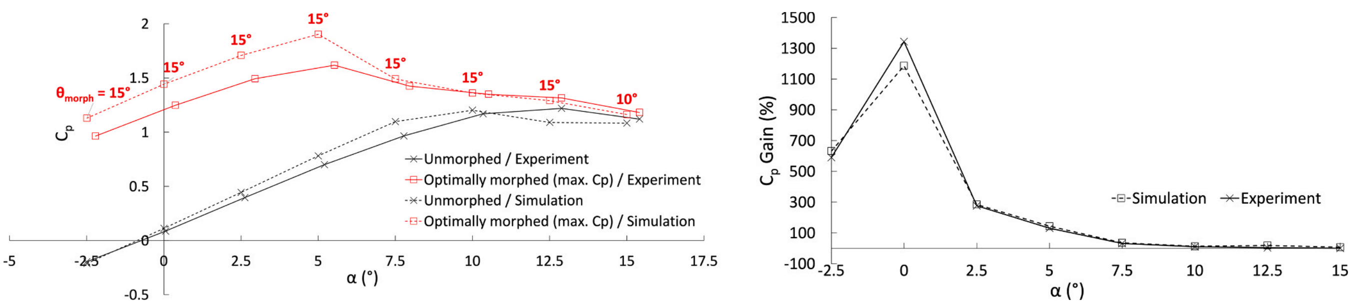


**Fig. 21.** Experimental validation of the morphing strategies for maximizing the power generated by the blade: Aerodynamic power coefficient (left) and associated gain (right) exhibited by the optimally morphed blade section $S_4$ within a flow of Reynolds number $Re = 0.25\times10^6$. Comparison of the experimental (lines) and computational (dashes) results.

chord, whose length is set to a maximal value of $c = 182.5$ mm. Notably, when combined with a flow velocity $U_\infty = 20$ m/s, this chord length value entails a Reynolds number $Re = 0.25\times10^6$, thereby corresponding exactly to the slowest flow regime that was explored numerically. The airfoil spans the test section almost entirely, with just a small gap (of 2.5 mm) left between the airfoil's tip and the vein's bottom wall. The latter gap is deemed small enough for not questioning the two-dimensional character of the flow (Barlow et al., 1999), which is thus assumed to be consistent with that of the 2D simulations. Finally, the airfoil is connected by its other tip to a load cell, namely a 6-axis force/torque transducer (brand *ATI Mini40*) capable of measuring loads of up to 80 N in both axial and lateral directions, with a resolution of 1/50 N. This load cell is attached to a motor-driven, rotating plate allowing to pitch the airfoil with a high accuracy (angular resolution of 0.01°).

### 5.2. Experimental acquisition and uncertainties

Each model is tested under the same flow conditions, namely, a velocity of $U_\infty = 20$ m/s and 8 incidences ($\alpha = -2.5°$ to 15°, by increments of 2.5°), thereby allowing a straight comparison against the simulation results ($Re = 0.25\times10^6$). The aerodynamic forces are acquired during 1 s at a sampling rate of 1 *kHz*, this being repeated three times in a row for consistency' sake.

Specific efforts are made to mitigate the various uncertainties weighting on the experiment, whether they come from the facility itself, the airfoil mock-ups, or the data acquisition process. For instance, a major source of uncertainty to be expected for such a typical closed-vein wind tunnel is the flow confinement and subsequent blockage effects, which may substantially alter the airfoil's aerodynamics. Using theoretical means (Barlow et al., 1999), this blockage effect is here estimated as rather moderate (of less than 10%, overall), being however corrected *a posteriori* through a semi-empirical correction of the raw aerodynamic data (Barlow et al., 1999). Another experimental bias may originate from the improper installation of the airfoil within the wind tunnel, namely its possible misalignment with respect to the flow direction. The cumulative offsets between the model, load cell, and flow orientations are thus checked prior to each test, being then corrected whenever needed. This verification step is conducted using various means, e.g., by aligning the airfoil with the wind tunnel centerline using two laser sheet projectors (see Fig. 19), by exerting the airfoil a predetermined load of specific orientation whereas checking the measured force. Besides, special attention is also paid to the dynamic effects incurred by the airfoil's incidence automation (Hu et al., 2007), which is here performed using a motorized rotating stage. Finally, a last possible source of error lies in all aeroelastic phenomena (e.g., flutter) to possibly occur during the test, especially considering the material used for the airfoil mocks-up (resin). It was checked that these phenomena were not triggered, except for a particular case, namely the morphed airfoil of maximum morphing ($\theta_{morph} = 15°$) set at the highest incidence ($\alpha = 15°$). In this particular case, the airfoil (free-end) tip was observed to vibrate, due to some probable flutter. To mitigate this, the corresponding data points were acquired with a higher sampling rate applied (3 *kHz*), thereby minimizing the risk of bias on the aerodynamic forces estimation (time averaging).

### 5.3. Experimental results and analysis

Fig. 20 depicts the lift and drag polars associated with all five airfoils, as tested under the present low speed flow regime. These curves can be compared with their computational counterpart of same Reynolds number ($Re = 0.25\times10^6$) – cf. top of Fig. 8. As can be seen, the agreement between both results is fairly good from both a qualitative and a quantitative viewpoint, especially for what concerns the lift coefficient. The agreement on the drag coefficient is a bit less satisfactory, with experimental values that are slightly higher than their computational counterparts, especially at low incidences ($\alpha < 7.5°$). These slight mismatches in drag can either be attributed to experimental biases (e.g., insufficient correction of blockage effects), or – more likely – to computational uncertainties. Indeed, it is notoriously challenging for steady RANS methods and their underlying turbulence models to capture accurately friction drag, especially when simulating low Reynolds number flows (for which viscous effects are more important) using a commercial solver (as done here). Aside from that, there is a slight but expected mismatch in lift and drag for that specific configuration which suffered from flutter ($\alpha = 15^\circ, \theta_{morph} = 15^\circ$), and whose results were thus purposely dashed in Fig. 20.

Despite these slight mismatches, the agreement between the experimental and computational results is good, overall. In particular, considering the optimal morphing strategy for a maximum power (cf. §4.2), the $C_P$ coefficient derived from the experimental data compares favourably with that of their computational counterpart. As an illustration, left side of Fig. 21 depicts the $C_P$ polars associated with both the baseline (unmorphed) and the optimally morphed airfoil, as inferred from either the experiment or the corresponding computations (whose results are reproduced from top/left of Fig. 12, for comparison purpose). As can be seen, the trend initially drawn through simulation is here globally recovered, namely a power coefficient by the morphed airfoil that is kept relatively constant across all incidences ($C_P \sim 1$–$1.5$), whereas being substantially magnified at low-to-moderate incidences ($\alpha \leq 7.5^\circ$). This is made even more explicit on the right side of the same Fig. 21, which compares the gain in $C_P$, as obtained from either the experiment or its computational counterpart (whose results are reproduced from Fig. 13, to ease the comparison). As can be seen, the experimental observations match closely the computational ones, thereby confirming the important $C_P$ benefits offered by the optimal morphing strategy, as well as the investigation method employed to characterize them.

This agreement between the computational and experimental results not only validates further the methodological approach used for the numerical investigation, but – more importantly – it confirms the phenomenological observations that were inferred from it. In particular, it confirms the substantial gains in aerodynamic power one can reach through a tailored morphing of blade sections. All this speaks in favor of further exploring the concept of blade morphing as a means to improve the performances of wind turbines.

## 6. Conclusion and perspectives

Regarding the further development of wind energy solutions, this study explored the idea of morphing wind turbine blades as a means to improve their aero-structural performances (increased aerodynamic power, reduced aerodynamic loads). To this end, the generic section of an existing wind turbine blade was morphed, with its trailing edge being deflected either downward or upward. The aerodynamic characteristics of the resulting morphed blade section were then assessed, this being done for a significant range of flow conditions (speed and incidence), with various levels of morphing applied. This was conducted through an extensive computational campaign relying on a Computational Fluid Dynamic (CFD) approach, which was first validated against reference data from the literature. Upon the numerical results obtained, various morphing scenarios were inferred, which sought at either alleviating the aero-structural loads exerted on the blade, or at rather maximizing its generated aerodynamic power. These optimal morphing strategies were shown to translate into significant benefits, among which substantial gains of power delivered by the blade section across all flow regimes. The phenomenology behind the benefits brought by the morphing was then explored, the effects of the airfoil shape deformation being discriminated from those of its sole chord line deflection (i.e., altered flow incidence). The former effects were shown to account for as much as the latter, which somehow demonstrates further that morphing a blade is superior to simply pitching or twisting it. Finally, the outcomes inferred from the computations were replicated experimentally through dedicated, small-scale tests, thereby further confirming the merits offered by the morphing.

Despite its exploratory character, the present study showcases the potential offered by morphable wind turbine blades. Going further in this exploration would require assessing more extensively the present concept of morphable blade, whether this concerns its structural aspects (overall design), its aerodynamic features (optimal morphing), or its subsequent aero-structural behavior. Concerning structural aspects, a natural follow-up would be to design a morphable structure (articulated skeleton, deformable skin, actuation devices, etc.) enabling to build an adaptive blade offering enough shape versatility against minimal structural penalties (weight, complexity). Regarding aerodynamic aspects, an immediate next step would be to incorporate more realism in the CFD computations by simulating a rotating, full 3D blade (Lanzafame et al., 2013; Jiang et al., 2022), each section of which would be morphed upon its local aerodynamic conditions. Ultimately, it would be needed to assess the overall aero-structural merits of the morphable blade, so that it guarantees the best trade-off between power efficiency (aerodynamic performances) and structural penalties (complexity, weight, etc.). To do so, use shall be made of advanced numerical simulations pertaining to aero-structural aspects (Li et al., 2020), namely Fluid-Structure Interaction (FSI). All the above prospects are currently investigated within an on-going study, which constitutes a direct continuation to the present one (Fong et al., 2024).

**Declaration of Competing Interest**

The authors declare that no conflict of interest exists.

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## Appendix A. Aerodynamics Characteristics and Power Efficiency of Wind Turbine Blades

Let us consider a wind turbine blade of radius R that rotates at an angular speed ($\omega$) due to an incoming flow of freestream velocity ($\mathbf{v}_\infty$) – cf. Fig. 1. The incoming flow speed (resp. blade's rotational speed) can be more universally characterized through the so-called Reynolds number, $Re_\infty$ (resp. Tip-Speed-Ratio, $TSR$). Taking the blade radius ($R$) and wind speed ($v_\infty = |\mathbf{v}_\infty|$) as characteristic quantities, these dimensionless parameters are respectively given by

$$Re_\infty \stackrel{\text{def}}{=} \frac{\rho v_\infty R}{\mu} \tag{A.1}$$

and

$$TSR \stackrel{\text{def}}{=} \frac{R\omega}{v_\infty} \tag{A.2}$$

### A.1 Aerodynamic Characteristics

For each blade section, the aerodynamic forces can be expressed as a function of the local airflow speed ($v_{tot.} = |\boldsymbol{v}_{tot.}|$) and lifting area (per elemental length) $S$, through the lift and drag coefficients, namely $C_L$and $C_D$;

$$|\boldsymbol{L}| = \frac{1}{2}\rho S C_L(\alpha) v_{tot.}^2 \tag{A.3}$$

and

$$|\boldsymbol{D}| = \frac{1}{2}\rho S C_D(\alpha) v_{tot.}^2 \tag{A.4}$$

These aerodynamic coefficients ($C_L$, $C_D$) are driven by both the airfoil characteristics (e.g., shape, chord length) and the airflow features, namely its incidence ($\alpha_{tot.}$) and – to a lesser extent – its speed ($v_{tot.}$). Taking the airfoil chord length ($c$) and airflow speed ($v_{tot.}$) as characteristic quantities, one can define the *local* Reynolds number ($Re$) specific to each blade section as follows:

$$Re = \frac{\rho v_{tot.} c}{\mu} \tag{A.5}$$

In all the above, $\rho$ and $\mu$ stand for the fluid density and viscosity, respectively.

Also specific to each blade section, the airfoil angle of attack ($\alpha$) results from the combination of the airflow incidence ($\alpha_{tot.}$) and the section's twist angle ($\beta$)

$$\alpha = \alpha_{tot.} - \beta \tag{A.6}$$

with

$$\beta = \beta_b + \beta_t \tag{A.7}$$

where $\beta_b$ is the (global) pitch angle of the blade, and $\beta_t$ the (local) twist angle of its section.

Besides, the airflow velocity vector ($\boldsymbol{v}_{tot.}$) results from the combination of the wind ($\boldsymbol{v}_\infty$) and blade advancement ($\boldsymbol{v}_\theta$) ones.

$$\boldsymbol{v}_{tot.} = \boldsymbol{v}_\infty - \boldsymbol{v}_\theta \tag{A.8}$$

Notably, the advancement velocity depends on both the rotational rate (or angular speed) of the blade ($\omega$) and the distance of the blade' section to its root ($r$)

$$\boldsymbol{v}_\theta = r\omega \boldsymbol{e}_\theta \tag{A.9}$$

In the previous equation, $\boldsymbol{e}_\theta$ is the unitary vector oriented in the advancement direction (corresponding to the $x$-axis in the sketch of Fig. 1). Of note, the rotational rate corresponds to the number of revolutions over a time length, which can be expressed as Revolutions Per Minute (RPM) or, as here, in radians per second (rad/s)

$$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T} \tag{A.10}$$

where $T$ is the temporal period for one rotation cycle.

### A.2 Power Characteristics

The power ($P$) extracted from the airflow by the wind turbine blade is classically defined as

$$P = (\boldsymbol{L} + \boldsymbol{D}) \cdot \boldsymbol{v}_\theta \tag{A.11}$$

It is easy to show that the projection on the $x$-axis gives

$$\boldsymbol{L} \cdot \boldsymbol{e}_\theta = |\boldsymbol{L}| \sin(\alpha_{tot.}) \tag{A.12}$$

and

$$\boldsymbol{D} \cdot \boldsymbol{e}_\theta = -|\boldsymbol{D}| \cos(\alpha_{tot.}) \tag{A.13}$$

By combining Eq. (A.3), (A.4), and (A.11- A.13), one gets

$$P = \left(\frac{1}{2}\rho S\right) v_{tot.}^2 [C_L(\alpha)\sin(\alpha_{tot.}) - C_D(\alpha)\cos(\alpha_{tot.})] v_\theta \tag{A.14}$$

with, upon Eq. (A.8),

$$v_{tot.} = |\boldsymbol{v}_{tot.}| = \sqrt{v_\infty^2 + v_\theta^2} \tag{A.15}$$

and, upon Eq. (A.9),

$$v_\theta = |\boldsymbol{v}_\theta| = r\omega \tag{A.16}$$

Noticing also that (cf. Fig. 1)

$$v_\infty = v_{tot.} \sin(\alpha_{tot.}) \tag{A.17}$$

and

$$v_\theta = v_{tot.}\cos(\alpha_{tot.}) \quad (A.18)$$

one can re-express Eq. (A.14) as

$$P = \left(\frac{1}{2}\rho S\right) v_{tot.}[C_L(\alpha)v_\infty - C_D(\alpha)v_\theta]v_\theta \quad (A.19)$$

Introducing now the ratio between the tip and wind speeds ($\lambda$)

$$\lambda = \frac{v_\theta}{v_\infty} \quad (A.20)$$

one immediately gets

$$v_{tot.} = v_\infty\sqrt{1+\lambda^2} \quad (A.21)$$

which leads to

$$P = \left(\frac{1}{2}\rho S\right) C_L(\alpha)v_\infty^2 v_\theta\sqrt{1+\lambda^2}\left[1 - \frac{C_D(\alpha)}{C_L(\alpha)}\lambda\quad\right] \quad (A.22)$$

By definition, the power coefficient $C_P$ is such that

$$P \stackrel{\text{def}}{=} \frac{1}{2}\rho S C_P v_\infty^3 \quad (A.23)$$

Therefore

$$C_P(\alpha,\lambda) = C_L(\alpha)\left\{\lambda\sqrt{1+\lambda^2}\left[1 - \frac{C_D(\alpha)}{C_L(\alpha)}\lambda\quad\right]\right\} \quad (A.24)$$

In summary, the power coefficient is proportional to the lift coefficient ($C_L$), with a proportionality factor that is driven by both the advancement-to-wind speeds ratio ($\lambda$) and the lift-to-drag ratio ($C_L/C_D$).

Notably, the advancement-to-wind speeds ratio ($\lambda$) not only depends on the wind force ($v_\infty$) and subsequent rotational frequency of the blade ($\omega$), but it is also driven by the location of the considered section along the span ($r$);

$$\lambda = \frac{r\omega}{v_\infty} \quad (A.25)$$

It can be re-expressed in terms of the Tip-Speed-Ratio number as follows

$$\lambda = \frac{r}{R}TSR \quad (A.26)$$

This ratio λ will be minimum at the root and maximal at the tip ($\lambda = TSR$).

On another hand, the aerodynamic coefficients ($C_L, C_D$) and their ratio depend on both the section's Reynolds number ($Re$, which is driven by the flow speed, $v_{tot.}$) and angle of attack ($\alpha$, which is driven by the flow orientation, $\alpha_{tot.}$). Notably, not only the airflow speed can be expressed in terms of the advancement-to-wind speeds ratio (cf. Eq. (A.21)), but also the airflow orientation. Indeed, by Eq. (A.17) and (A.18), one gets

$$\alpha_{tot.} = \mathrm{Atan}\left(\frac{1}{\lambda}\right) \quad (A.27)$$

Therefore, the blade section's local Reynolds number can be recast into

$$Re = \sqrt{1+\lambda^2}\left(\frac{c}{R}\right)Re_\infty \quad (A.28)$$

where $Re_\infty$ stands for the overall Reynolds number of the entire wind turbine blade under a given wind of freestream speed ($v_\infty$), cf. Eq. (A.1).

In summary, for a given blade rotating with a given $TSR$ due to an incoming flow of Reynolds number $Re_\infty$, one can describe the local aerodynamic ($C_L, C_D$, $C_L/C_D$) and power ($C_P$) characteristics associated with each blade section, all of them being driven by the ratio $r/R$ (through the $\lambda \Leftrightarrow TSR$ relationship) and $c/R$ (through the $Re \Leftrightarrow Re_\infty$ relationship).

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